(Please write your Enrollment Number) **Enrollment No.** 06618103 2025

MID-TERM EXAMINATION

(Programme: B. Tech. AI&ML / IT) (Semester: II)

(March 2026) OFF LINE mode

Subject Code: BIT 102	**Subject: Object Oriented Programming**
Time : 1 ½Hours	**Maximum Marks :30**

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		**(2.5*4=10)**	**CO Mapping**
	a) Differentiate between Reference and Pointer in C++ with suitable examples.		CO1
	b) Consider the following code:		CO1

```
#include<iostream>
using namespace std;
int main(){
        int a[5] = {7,5,76,9,54};
        int *p;
        p=a;
        cout<<*p<<endl;
        cout<<*p++<<endl;
        cout<<*(++p);
}
```

What is the output of the above program?

	c) What is the use of static member variable in OOP? Explain with suitable example.		CO2
	d) Consider the following code:		CO2

```
#include <iostream>
using namespace std;
class A {
    public:
        A() { cout << "Constructor\n"; }
        ~A() { cout << "Destructor\n"; }
};
int main() {
    A obj1;
    {
        A obj2;
    }
    return 0;
}
```

What is the output of the above program? Also explain the steps of execution.

	UNIT I		**CO Mapping**
Q2	**Attempt any two parts**	**(5*2=10)**	
	a) Are Encapsulation and Data Abstraction the same concepts in OOP? Justify your answer with suitable example.	5	CO1
	b) What is the use of inline function in C++? How is it different from a function with default arguments? Explain with suitable examples.	5	CO1
	c) Write a C++ program to print whether a given number is palindrome or not.	5	CO1
	UNIT II		**CO Mapping**
Q3	**Attempt any two parts**	**(5*2=10)**	
	a) Write a C++ program to overload the unary increment operator (++) for both prefix and postfix forms.	5	CO2
	b) What is a constructor? Differentiate between default and parameterized constructor with suitable example.	5	CO2
	c) Write a C++ program to implement a class EMPLOYEE having the following members: 1. Member variables: • Name of employee • Salary • Static variable to count total number of employees 2. Member functions: • assign() – to assign employee details • display() – to display employee details • showCount() – to display total number of employees Use appropriate access specifiers and data types to justify the output of the program.	5	CO2